

Section A

Answer **all** questions in this section in the spaces provided.

- 1(a)** For decades, scientists were puzzled by why dinosaurs suddenly became extinct 65 million years ago. In studying core samples of rock dating back to that period, scientists found an unusual high level of iridium. This possibly came from an iridium-rich asteroid that struck the Earth's surface and the mystery to dinosaurs' extinction was solved.

Iridium exists as two naturally occurring isotopes, iridium-191 and iridium-193.

- (i)** Define the term *relative atomic mass*.

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- (ii)** The relative atomic mass of iridium is 192.2. Calculate the natural abundance, in percentage, of each isotope.

- (iii)** Iridium is one of the rarest metals in the Earth's crust. It is thought to exist in 2 parts per billion (ppb). Calculate the mass of iridium (in kg), in 10 tonnes of Earth's crust.

[1 billion = 10^9 , 1 tonne = 1000 kg]

[4]

(b) Aqua regia (a mixture of 75% nitric acid and 25% hydrochloric acid by volume) is highly corrosive. Only noble metals like iridium are inert to this solution. A 5 g sample of platinum-iridium alloy required 24.6 cm^3 of aqua regia for complete reaction. Platinum was completely oxidised to platinum(IV) ions by nitric acid and 0.5 g of metal was recovered.

(i) Find the percentage of each metal in the alloy.

(ii) The concentration of nitric acid used to make aqua regia is 5.0 mol dm^{-3} . Assuming that the reaction between the sample and aqua regia is complete, determine the amount of HNO_3 and Pt used in the reaction.

(iii) Hence construct a balanced equation for the reaction between HNO_3 and Pt.

[4]

(c) Man-made iridium-192 is an important radioactive isotope of iridium. It has a half-life of 74 days. The decay of a sample of iridium-192 was monitored.

(i) Calculate the original mass of the sample if the sample weighed 0.05 g after 222 days.

(ii) Suggest an application for iridium-192.

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[2]

[Total: 10]

- 2 **Figure 1** shows the first eight ionisation energies of an element **X** while **Figure 2** shows the first ionisation energies of the eleven consecutive elements (including element **X**) from Period 2 and Period 3.

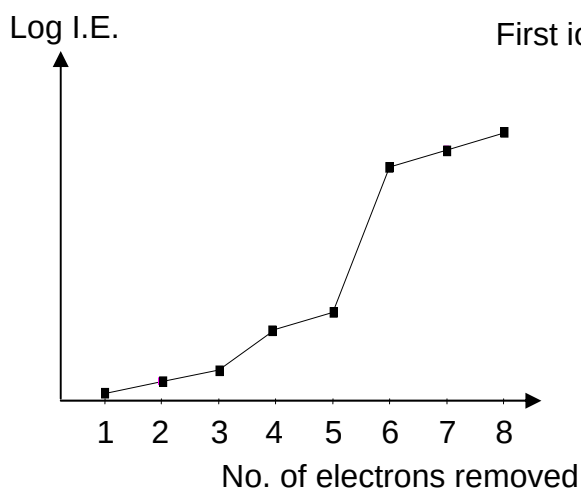


Figure 1

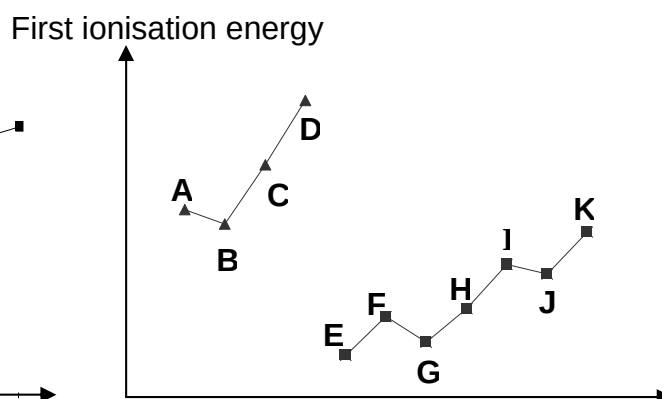


Figure 2

- (a) (i) With reference to **Figure 1**, explain briefly the group to which element **X** belongs. Hence, identify element **X**.

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- (ii) Which letter in **Figure 2** corresponds to element **X**? Explain your answer briefly.

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- (iii) Write the electronic configuration of element **X** and draw two labelled diagrams of the two p orbitals of element **X** to illustrate their shapes and sizes.

[6]

- (b) (i) Chlorides of elements **F** and **H** react with water separately to give solutions of pH values 6.5 and 2 respectively. Identify **F** and **H**. Hence account for the difference in pH values, with the help of equations

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- (ii) The acid-base nature of oxides of **E** and **I** are different. Illustrate this property by writing balanced equations to show their reactions, if any, with aqueous sodium hydroxide and aqueous hydrochloric acid.

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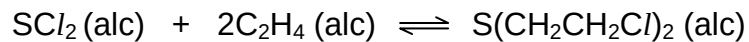
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[4]

[Total: 10]

- 3 (a) A United Nations toxicologist studying the properties of mustard gas, $\text{S}(\text{CH}_2\text{CH}_2\text{Cl})_2$, a blistering agent used in warfare, mixes 50 cm^3 of 1.34 mol dm^{-3} SCl_2 with 50 cm^3 of 1.96 mol dm^{-3} C_2H_4 , and allows it to react at 20°C . At equilibrium, the concentration of $\text{S}(\text{CH}_2\text{CH}_2\text{Cl})_2$ is found to be $0.350 \text{ mol dm}^{-3}$.



Give the expression for K_c and calculate its value for the above reaction at 20°C .

[3]

- (b) Explain the terms *order of reaction* and *half-life*.

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[2]

- (c) Hydrogen peroxide reacts with acidified iodide ions to give iodine. The following results were obtained in a study of the kinetics of this reaction.

Experiment	Initial concentrations of reactants/ mol dm ⁻³			Initial rate of formation of iodine/ mol dm ⁻³ s ⁻¹
	[I ⁻]	[H ₂ O ₂]	[H ⁺]	
1	0.010	0.020	0.10	6.0 × 10 ⁻⁶
2	0.010	0.030	0.20	9.0 × 10 ⁻⁶
3	0.020	0.030	0.10	1.8 × 10 ⁻⁵

- (i) Write a balanced equation for the reaction.

- (ii) The reaction is zero order with respect to hydrogen ions. Determine the order of reaction with respect to iodide and hydrogen peroxide.

Order of reaction with respect to iodide :

Working:

Order of reaction with respect to hydrogen peroxide :

Working:

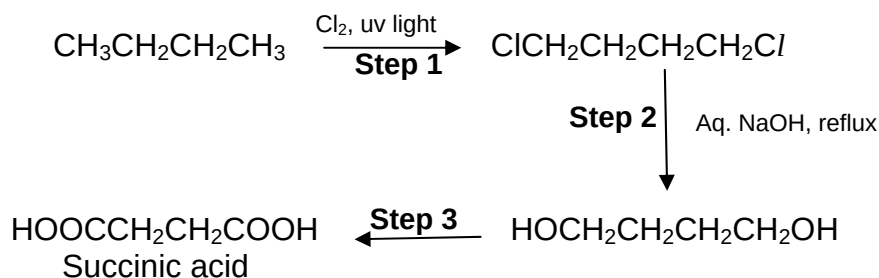
(iii) Write a rate equation for this reaction.

(iv) Hence, calculate a value for the rate constant, giving its units.

[5]

[Total: 10]

- 4 Succinic acid (butanedioic acid) is widely used in the food and beverage industry. Industrially it is produced from the fermentation of carbohydrates. A student proposes the following synthetic route for the synthesis of succinic acid from butane:



- (a) Suggest the reagents and conditions for **step 3**.

..... [1]

- (b) Name the type of reaction involved in the first step. Hence comment on the yield of acid obtained.

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[2]

- (c) Suggest an alternative route for synthesizing succinic acid starting from ethene. State clearly all reagents and conditions required.

[5]

- (d) Calculate the volume of gas evolved at rtp when an excess of sodium metal is added to 60 g of the succinic acid.

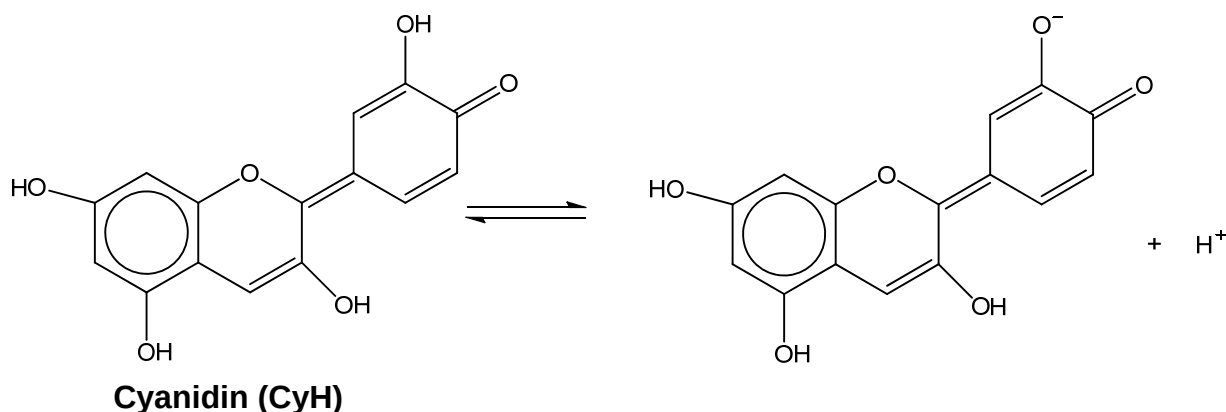
[2]

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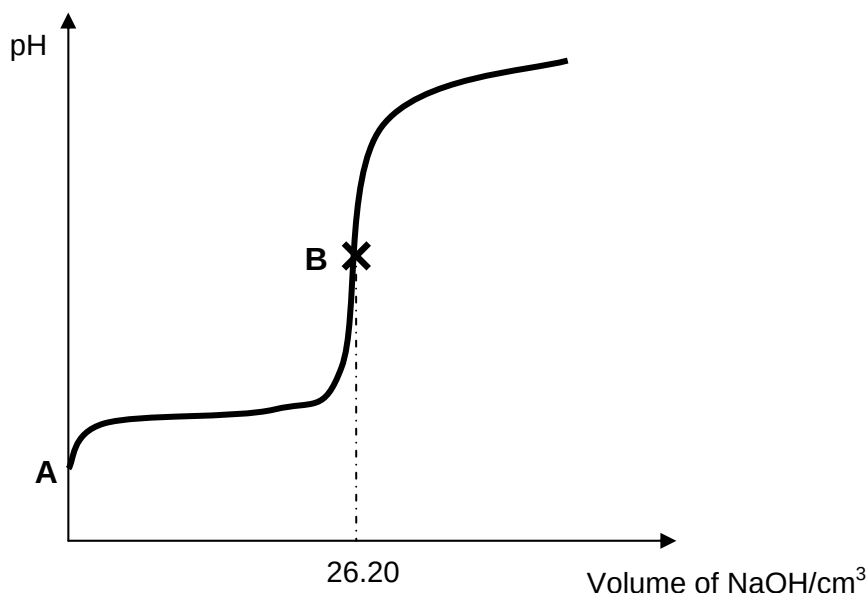
Section B

Answer **two** of the three questions in this section on separate answer paper.

- 5 Cyanidin is a natural pH indicator that can be found in many red berries, including grapes, blackberry and cherry. Aqueous solutions of cyanidin changes colour with pH, appearing red at $\text{pH} < 3$, violet at $\text{pH} 7-8$, and blue at $\text{pH} > 11$. In addition, cyanidins are antioxidants that can scavenge free radicals and counteract ageing caused by oxidative damage in our body tissues and cells.



Cyanidin (CyH) is a *weak monoprotic acid*. In one experiment, 25.0 cm^3 of cyanidin was titrated against 0.10 mol dm^{-3} of dilute NaOH at 25°C . The following titration curve was obtained using a data logger.



- (a) (i) Explain what is meant by a *weak monoprotic acid*.
- (ii) Calculate the molar concentration of the cyanidin solution.
- (iii) Given that the solution of cyanidin is only 0.8% dissociated into ions, calculate the pH at point A.

(iv) Write an expression for the acid dissociation constant, K_a , for cyanidin. Hence, calculate the value of K_a for cyanidin, stating clearly its units.

(v) Explain, with the aid of an appropriate equation, why the pH at end-point **B** is greater than 7.

[7]

(b) An aqueous solution of cyanidin, CyH and Cy^- can act as a buffer.

(i) Copy the titration curve onto your answer script and label

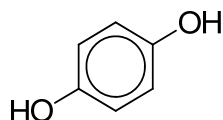
I – the buffer region

II – the point corresponding to maximum buffer capacity.

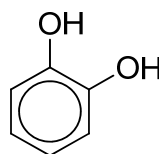
(ii) Explain, with the help of equations, how an aqueous mixture of CyH and Cy^- can control pH when relatively small amounts of acid or base is added to the solution.

[5]

(c) Cyanidin can be synthesized via a number of steps from either of the two molecules shown below:



Molecule P



Molecule Q

(i) Which molecule **P** or **Q**, will have a higher boiling point? Explain your answer.

(ii) **P** dissolves readily in water at room temperature but two immiscible layers were observed when **P** was added to trichloromethane. Account for the above observation.

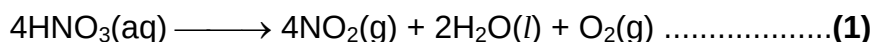
[5]

(d) Predict and explain the relative acidity of propanoic acid and 2-chloropropanoic acid.

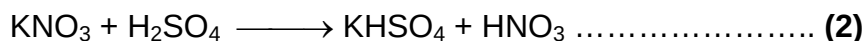
[3]

[Total: 20]

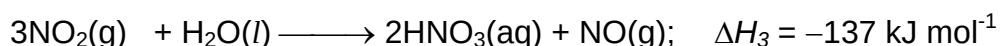
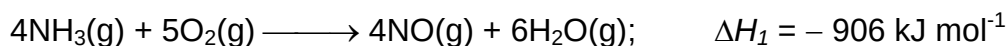
- 6 Nitric acid, HNO_3 , known to early chemists as aqua fortis (meaning "strong water"), is a major commercial acid. It is available as the concentrated acid which is a 70% aqueous solution. Although colourless when pure, the acid turns yellow when exposed to light as a result of decomposition.



Pure nitric acid can be prepared by distillation from a mixture of potassium nitrate and concentrated sulfuric acid as shown

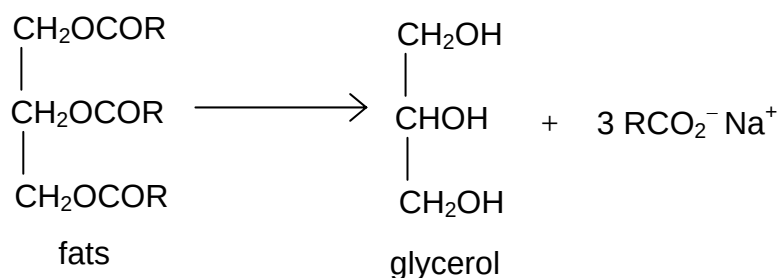


This method was at one time used in the commercial preparation of nitric acid, but it has now been replaced by the Ostwald process, discovered in 1902 by the German chemist Wilhelm Ostwald, who received the Nobel Prize in chemistry in 1909 for his work on catalysts. In the Ostwald process, ammonia gas is burned in oxygen over a platinum catalyst. The product is nitric oxide, NO , which reacts with oxygen to give nitrogen dioxide. When dissolved in water, NO_2 produces nitric acid. The equations are

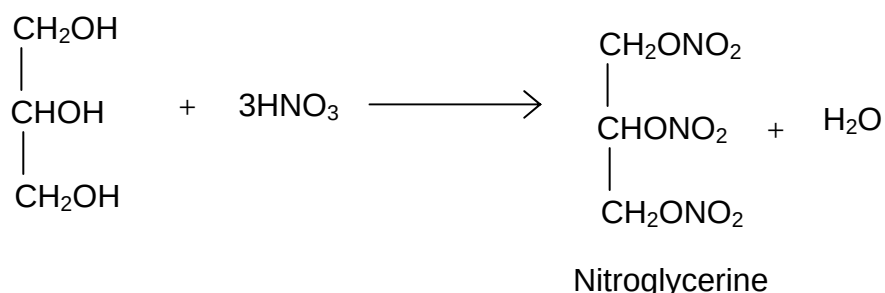


The largest single use of nitric acid is in the production of ammonium nitrate, NH_4NO_3 , which is largely used as an agricultural fertiliser. It is also used to manufacture chemicals for nylon and to make explosives like nitroglycerine, $\text{C}_3\text{H}_5(\text{NO}_3)_3$. In making the explosive nitroglycerine, glycerol, $\text{C}_3\text{H}_5(\text{OH})_3$, which is obtained from fats is used as the starting material. The manufacturing process is shown in two stages as shown below:

Stage 1



Stage 2

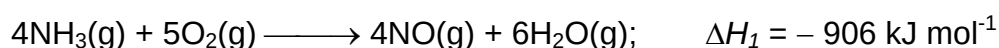


(a) Explain why nitric acid in a bottle often turns yellow after some time. [1]

(b) State the type of reaction involved in the making of pure nitric acid as exemplified in reaction (2). Explain your answer. [2]

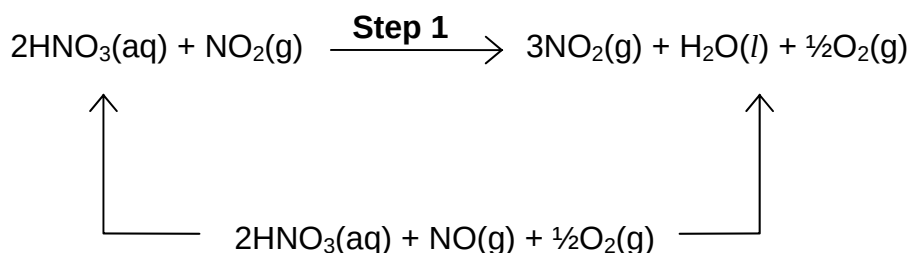
(c) Draw the dot and cross diagrams for the ions in NH_4NO_3 . Hence, predict the shape of each ion present. [4]

(d) (i) Using relevant data from the *Data Booklet*, calculate the average bond energy of the $\text{N}=\text{O}$ bond in the following reaction,



(ii) Suggest a reason for the discrepancy between your answer in part (d)(i) and the experimentally determined value of 629 kJ mol^{-1} . [4]

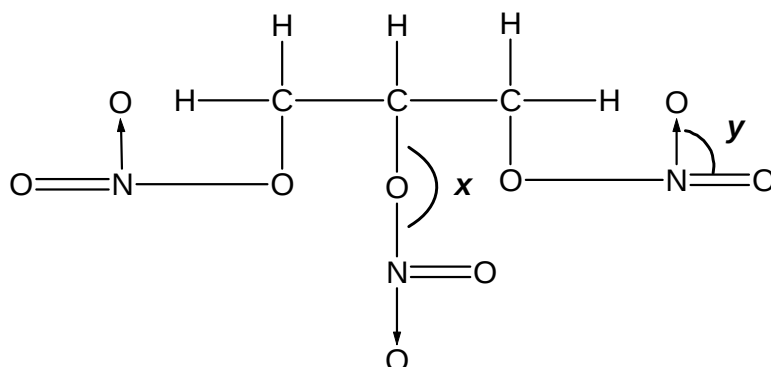
(e) Using appropriate enthalpy change values given for the Ostwald Process and the following energy cycle, calculate the enthalpy change for **Step 1**.



[2]

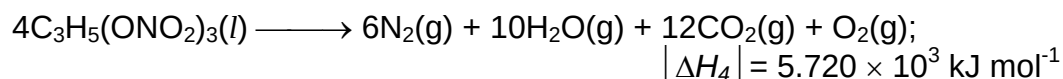
(f) State the reagents and conditions used for the formation of glycerol from fats. [1]

(g) The molecule of glycerin has the following structure:



State the $\text{C}-\text{O}-\text{N}$ bond angle (x) and the $\text{O}-\text{N}-\text{O}$ bond angle (y) as indicated in the diagram above. [2]

- (h) Nitroglycerine is hazardous to handle in the pure state. It is very sensitive to shock, as shown by the following reaction:



Alfred Nobel (1833 -1896) discovered that when nitroglycerine was mixed with clay and similar materials, it was much less shock-sensitive. The mixture is called dynamite. One stick of dynamite weighs 500 g and contains 80% of nitroglycerine by mass.

- (i) Predict the sign of ΔH_4 and explain your answer.
- (ii) Dynamite is still widely used in many countries for clearing of obstacles and for mining resources. However, its potential danger of explosion cannot be underestimated.

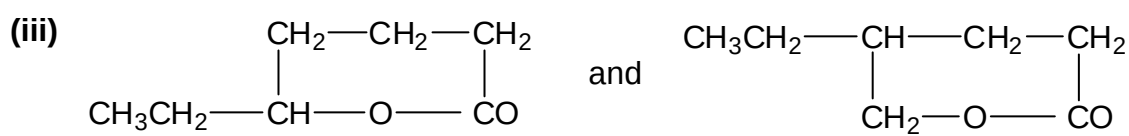
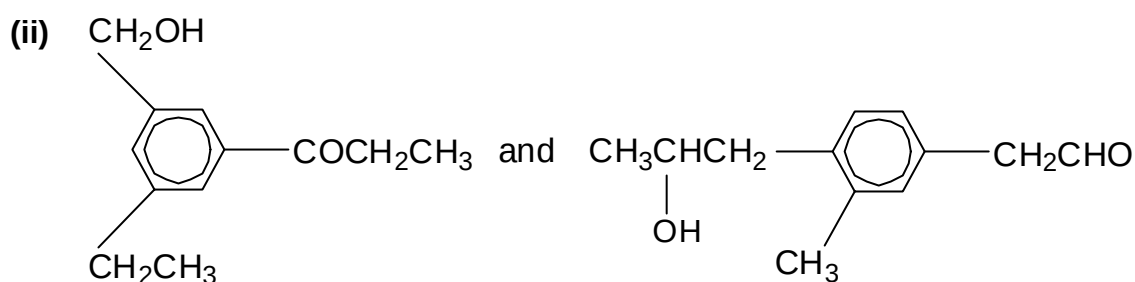
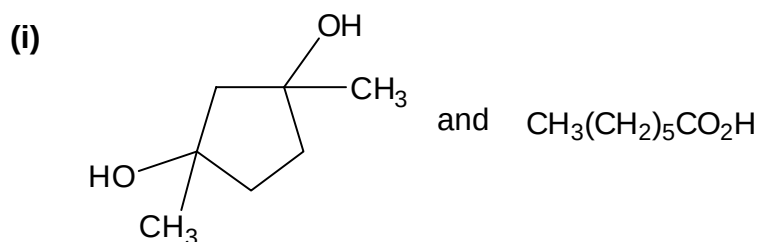
Calculate the average change in temperature of the surrounding 20 m^3 of air around **one** stick of exploding dynamite.

[Volumetric heat capacity of air, $C = 1.297 \text{ kJ m}^{-3} \text{ K}^{-1}$]

[4]

[Total: 20]

- 7 (a) Suggest methods by which the following pairs of compounds could be distinguished from each other by simple chemical test. State clearly how each compound would behave. For each case, **no more than two steps** should be used.



[7]

- (b)** **A** ($\text{C}_4\text{H}_8\text{Cl}_2$) reacts with sodium hydroxide under different conditions. In aqueous medium, **A** forms **B** ($\text{C}_4\text{H}_{10}\text{O}_2$) while in ethanolic medium, **A** forms the major product **C** ($\text{C}_4\text{H}_7\text{Cl}$). 1 mol of **C** reacts with 1 mol of aqueous bromine.

B reacts with warm alkaline aqueous iodine to give a pale yellow precipitate as well as **D** ($\text{C}_3\text{H}_5\text{O}_3\text{Na}$). Both **C** and **D** undergo vigorous oxidation in acidic medium to produce **E** and **F** respectively.

0.0500 mol of **F** was dissolved in water and the solution was made up to 250 cm³. A 25.0 cm³ portion of this solution required 13.30 cm³ of 0.75 mol dm⁻³ NaOH for complete reaction.

When **E** is refluxed with alcoholic KCN, the product formed has the molecular formula $C_3H_3NO_2$.

- (i) Deduce the structures of all the lettered compounds (**A** to **F**) and explain the reactions involved.
- (ii) Write an equation to represent the conversion of **A** to **C**.

[13]

[Total: 20]